SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I **N. Hema sree (192511100)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: N. Hema sree

Annexure A

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

Given:

- File size = **150 MB = 150 × 10^6 = 150,000,000 bytes**
- Segment size = **1500 bytes**
- Data Link overhead = **40 bytes/frame**
- Bandwidth = **50 Mbps**

Calculation

Number of segments
= 150,000,000 ÷ 1500
= **100,000 segments**

Number of frames
= **100,000 frames**

Total overhead
= 100,000 × 40
= **4,000,000 bytes (4 MB)**

Total data transmitted
= 150 MB + 4 MB
= **154 MB**

Transmission time
= 154 × 8 = **1232 Mb**

Time = 1232 ÷ 50
= **24.64 seconds**

Explanation

- The **Transport Layer** divides data into segments and ensures reliable delivery using acknowledgements and error recovery.
- The **Data Link Layer** adds headers/trailers, detects transmission errors using CRC, and ensures reliable frame delivery between adjacent devices.

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

Given

- Window size = **6**
- Total frames = **48**
- Frame size = **1024 bytes**

Calculation

Transmission windows
= 48 ÷ 6
= **8 windows**

Retransmissions
= 1 lost frame/window

= **8 retransmissions**

Explanation

Flow control prevents the sender from transmitting data faster than the receiver can process it, reducing congestion and improving efficiency.

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

Given

- Packet size = **12,000 bytes**
- MTU = **1500 bytes**
- Header = **20 bytes**

Calculation

Maximum data per fragment

= 1500 − 20

= **1480 bytes**

Fragments

= Ceiling(12000 ÷ 1480)

= **9 fragments**

Header overhead

= 9 × 20

= **180 bytes**

Explanation

The **Network Layer** fragments large packets to match the MTU and reassembles them at the destination using identification, fragment offset, and flags.

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

- Number of windows = **8**
- Retransmissions = **8**

Explanation

Flow control avoids buffer overflow and ensures smooth communication by controlling the sender's transmission rate.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

Given

- File = **600 MB**
- Checkpoint every **60 MB**
- Failure after **390 MB**

Calculation

Checkpoints created

390 ÷ 60 = 6.5

= **6 checkpoints**

Last checkpoint

= **360 MB**

Retransmission required

390 − 360

= **30 MB**

Explanation

Synchronization checkpoints allow transmission to resume from the last saved point instead of restarting the entire transfer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

Given

- Original file = **900 MB**
- Compression = **40%**
- Encryption overhead = **5%**

Calculation

Compressed size

= 900 × 60%

= **540 MB**

Encrypted size

= 540 × 1.05

= **567 MB**

Overall reduction

= ((900 − 567) ÷ 900) × 100

= **37% reduction**

Explanation

The **Presentation Layer** compresses data to reduce transmission time and encrypts it to ensure confidentiality.

7. An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

Given

- Data = **3 GB = 3000 MB**
- Request size = **3 MB**
- Throughput = **120 Mbps**

Calculation

Requests

3000 ÷ 3

= **1000 requests**

Transmission time

3000 MB × 8

= **24,000 Mb**

Time

= 24,000 ÷ 120

= **200 seconds**

Explanation

The **Application Layer** provides services such as FTP, manages file transfers, user authentication, and error handling.

8.A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

Given

- Application = 4 ms
- Transport = 3 ms
- Network = 5 ms
- Data Link = 2 ms

- Physical = 1 ms
- Propagation = 18 ms
- Transmission = 12 ms

Calculation

Total delay

= 4 + 3 + 5 + 2 + 1 + 18 + 12

= **45 ms**

Explanation

Each OSI layer performs a specific function such as application processing, segmentation, routing, framing, and physical transmission, contributing to the total communication delay.

9.A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers.

Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

Given

- Useful data = **80 MB**
- Frame size = **1600 bytes**
- Overhead = **80 bytes**

Calculation

Useful data

= 80,000,000 bytes

Frames

= 80,000,000 ÷ (1600 − 80)

= 80,000,000 ÷ 1520

≈ **52,632 frames**

Total overhead

= 52,632 × 80

= **4,210,560 bytes (≈4.21 MB)**

Transmission efficiency

= (1520 ÷ 1600) × 100

= **95%**

Explanation

Protocol overhead provides addressing and error control but slightly reduces the amount of useful data transmitted, affecting network efficiency.

10. A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

Given

- Original file = **240 MB**
- Compression = **30%**
- Segment size = **1400 bytes**
- Data Link overhead = **60 bytes**

Calculation

Compressed size

= 240 × 70%

= **168 MB**

Compressed bytes

= **168,000,000 bytes**

Segments

= 168,000,000 ÷ 1400

= **120,000 segments**

Protocol overhead

= 120,000 × 60

= **7,200,000 bytes (7.2 MB)**

Final transmitted data

= 168 MB + 7.2 MB

= **175.2 MB**

Explanation

- **Presentation Layer:** Compresses and encrypts data.
- **Transport Layer:** Segments the data and ensures reliable delivery.
- **Data Link Layer:** Adds frame headers/trailers and performs error detection.
- **Physical Layer:** Transmits the final data over the network.

These layers work together to provide secure, efficient, and reliable communication.

RUBRICS FOR CALCULATION

Criteria	Weighta g e	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understandi n g of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understandin g with some misconceptio n s	Poor understanding ; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation ; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization , incoherent answers
		and coherence			
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandab l e with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand